

Advanced RAG

C1. Multimodal RAG for Product Manuals

For this type of system, I would use multimodal RAG because the useful information is not always present in the text. The product manuals can contain both written instructions and visual information such as diagrams and photos.

First, I would extract the text as well as the images and diagrams from the manuals. The text can be converted into text embeddings, while the images can be represented using a multimodal or image embedding model. These representations can then be stored and used for retrieval.

When a user asks, “How do I attach the side panel?”, the query would be used to retrieve the relevant text and visual content from the manual. If the actual instructions are shown only in a diagram, the system should be able to retrieve that diagram and provide it to the multimodal language model. The model can then understand the diagram and explain the steps to the user.

If I used a text-only RAG pipeline, the diagram would either be ignored or only its surrounding text would be retrieved. In this case, the most important information about attaching the side panel could be completely missed. This could lead to an incomplete or incorrect answer even though the correct information exists in the manual.

C2. Unified Embedding Space vs Separate Indexes with Fusion

There are two common approaches for multimodal retrieval.

In a **unified embedding space**, text and images are converted into embeddings that exist in the same vector space. This allows a text query to directly search across both text and images using the same similarity measure.

With **separate indexes with fusion**, text and images are stored and searched separately. The text retrieval system searches the text index and the image retrieval system searches the image index. The results from both searches are then combined or ranked together.

For the product manual scenario, I would use the **separate indexes with fusion** approach. Product manuals can contain different types of information, and having separate indexes gives more control over how text and visual results are retrieved. A question about attaching a side panel could retrieve relevant written instructions from the text index and the corresponding assembly diagram from the image index. These results can then be combined and passed to the multimodal model.

This approach also makes it easier to handle each type of content using retrieval methods that work well for that particular modality.